



Project Charter

Fiberoptiq Distributed LiquidCool Infrastructure Network (DLIN) Program

A Decentralized High-Performance Compute and Energy-Efficient Digital Infrastructure Initiative

Executive / Project Summary

The Fiberoptiq Distributed LiquidCool Infrastructure Network (DLIN) Program is a decentralized data infrastructure initiative focused on repurposing qualified commercial, government, institutional, telecom, utility-connected, and energy-intensive facilities into energy-efficient distributed compute environments utilizing Fiberoptiq and LiquidCool Solutions submerged server technologies.

The project establishes a scalable proof-of-concept and operational business framework designed to reduce cooling-related power consumption, improve compute density, lower infrastructure operating costs, increase distributed network resiliency, and create monetizable digital infrastructure assets through strategic deployment partnerships, licensing, thermal recovery integration, and secure high-performance computing operations.

The initiative supports phased deployment across distributed edge and regional compute locations aligned with modern cloud, AI, HPC, telecom, cybersecurity, and digital infrastructure requirements while contributing positively toward infrastructure modernization, energy efficiency, and community digital connectivity.

Project Goals

- Deploy decentralized LiquidCool-enabled compute infrastructure across qualified facilities within a phased 24–36 month deployment timeline.
 - Reduce cooling-related energy consumption compared to traditional air-cooled data center environments.
 - Increase compute density and operational efficiency through submerged server technologies.
 - Establish secure and resilient distributed edge-compute infrastructure supporting government and enterprise workloads.
 - Develop strategic partnerships with government agencies, utilities, telecom operators, and commercial real estate stakeholders.
 - Enable scalable deployment through a capital-efficient licensing, royalty, and equity participation model.
 - Implement measurable thermal recovery and energy optimization systems where operationally feasible.
 - Support AI, cloud, HPC, blockchain, and secure enterprise compute services through distributed infrastructure deployment.
 - Create recurring revenue opportunities through infrastructure hosting, compute leasing, and strategic service agreements.
 - Establish an operational proof-of-concept demonstrating the viability of decentralized energy-efficient digital infrastructure.
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Deliverables

- Distributed LiquidCool infrastructure deployment framework
 - Pilot-site qualification and deployment roadmap
 - Installed submerged compute server infrastructure
 - Distributed network connectivity architecture
 - Cybersecurity and compliance integration framework
 - Operational monitoring and telemetry systems
 - Energy-efficiency benchmarking and reporting metrics
 - Strategic partnership and stakeholder engagement model
 - Licensing and deployment agreement framework
 - Proof-of-concept operational deployment and performance validation
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Business Case / Background

- Traditional data centers face rising energy consumption and cooling costs.
 - Centralized hyperscale infrastructure creates operational and geographic concentration risks.
 - Existing commercial and institutional facilities represent underutilized infrastructure assets.
 - LiquidCool submerged server technology improves thermal efficiency and compute density.
 - Distributed edge-compute demand continues to increase due to AI, cloud, and digital-service growth.
 - Government and enterprise sectors require resilient and secure digital infrastructure modernization.
 - Energy optimization and thermal recovery initiatives align with ESG and sustainability objectives.
 - Distributed infrastructure lowers deployment barriers compared to large greenfield data center construction.
 - The licensed 64-patent portfolio supports differentiated market positioning and deployment scalability.
 - Strategic partnerships create opportunities for long-term recurring infrastructure revenue streams.
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Benefits and Cost

Benefits

- Reduced cooling and operational energy consumption
- Lower infrastructure operating costs per compute unit
- Increased compute density and deployment flexibility
- Improved regional infrastructure resiliency and redundancy
- Repurposing of existing commercial infrastructure assets
- Enhanced secure connectivity for enterprise and government workloads
- Potential thermal recovery and energy optimization value creation
- Scalable recurring revenue through compute and infrastructure services
- Alignment with sustainability and infrastructure modernization initiatives

Cost Considerations

- Facility retrofit and infrastructure conversion costs
 - Submerged server hardware and deployment equipment
 - Network, power, and cybersecurity integration costs
 - Compliance, permitting, and engineering assessments
 - Staffing, operational readiness, and maintenance support
 - Pilot deployment and phased expansion capital requirements
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Scope and Exclusion

In Scope

- Deployment of Fiberoptiq and LiquidCool infrastructure technologies
- Conversion of approved facilities into distributed compute nodes
- Network, power, and operational systems integration
- Cybersecurity and compliance implementation
- Energy-efficiency and thermal recovery integration
- Government, utility, telecom, and commercial partnership development
- Pilot deployments and phased regional expansion

Out of Scope

- Construction of large centralized hyperscale campuses
 - Semiconductor manufacturing or hardware fabrication ownership
 - Non-aligned speculative technology ventures outside approved infrastructure objectives
 - Utility-scale power generation ownership unrelated to deployment operations
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Project Sponsor

Director of Operations

Project Lead

Project Manager

Project Team

- Infrastructure Delivery Manager
- Technical Program Manager (TPM) – Network/Cloud
- IT Risk & Compliance Manager
- Operations Director – Data Center / Edge Compute
- Vendor Management Office (VMO) Lead

Measuring Success – What Is Acceptable

- Successful deployment of initial pilot compute sites within approved timelines
 - Measurable reduction in cooling-related energy consumption
 - Stable operational uptime and network resiliency benchmarks achieved
 - Successful integration of submerged server technologies into distributed facilities
 - Positive infrastructure performance validation against traditional data center benchmarks
 - Acquisition of strategic government, utility, telecom, or enterprise partnerships
 - Achievement of operational cybersecurity and compliance standards
 - Demonstrated scalability of distributed compute deployment model
 - Establishment of recurring revenue-generating infrastructure services
 - Validation of proof-of-concept business and infrastructure model viability
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Budget Needed – What Is Acceptable

Estimated Pilot Phase Budget Range

Phase 1–2 Proof-of-Concept / Initial Deployment:

\$5 Million – \$25 Million USD

Dependent upon:

- Number of pilot facilities
- Power and retrofit requirements
- Compute deployment scale
- Network and cybersecurity architecture
- Thermal recovery integration scope
- Staffing and operational readiness

Estimated Regional Expansion Budget Range

Multi-Site Distributed Infrastructure Expansion:

\$50 Million – \$250 Million+ USD

Dependent upon:

- Regional deployment scale
 - Government and enterprise contract acquisition
 - Infrastructure density and compute demand
 - Utility integration and energy optimization requirements
 - National or multi-regional deployment strategy
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Appendix: Misalignments and Decisions

Potential Misalignments

- Expansion beyond approved SMART-goal-driven deployment objectives
- Overextension into unrelated technology sectors
- Insufficient facility power or connectivity requirements
- Regulatory or compliance limitations by deployment region
- Capital deployment exceeding phased operational readiness
- Misalignment between stakeholder expectations and operational capabilities

Strategic Decisions

- Maintain decentralized infrastructure deployment strategy
- Prioritize energy-efficient and scalable compute environments
- Utilize phased pilot deployment before aggressive expansion
- Focus on strategic partnerships over direct infrastructure ownership expansion
- Align all deployment activity with measurable SMART goals and project scope boundaries
- Maintain compliance-focused operational governance throughout deployment lifecycle